# Next-Day SF Traffic Forecaster

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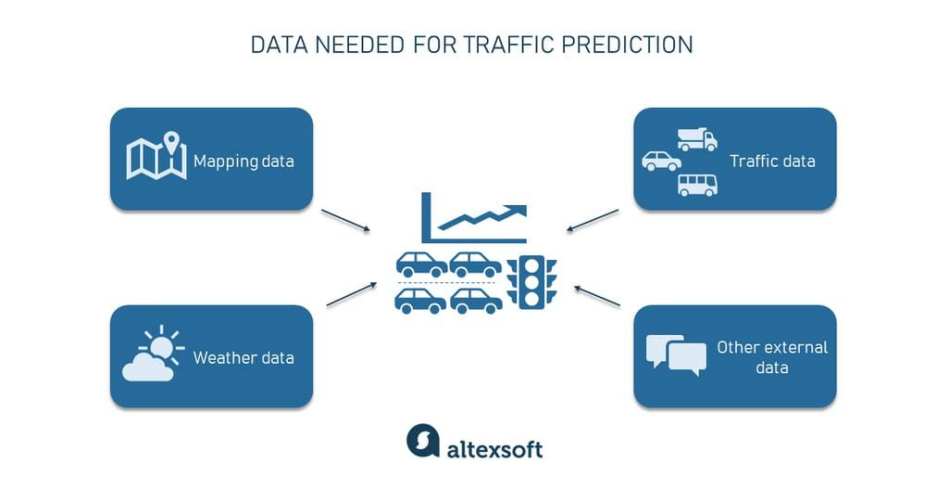
# Problem

As of right now people in San Francisco and everywhere in the world are stuck with heavy traffic and most of them only find out how bad it is once they are on the road. San Francisco is also ranked the third worst city for traffic congestion. Apps like google maps and waze only show the current traffic condition and can’t help people plan ahead. Especially when it comes to being at school or work on time people wouldn’t know what to expect of traffic unless they are already there. A lot of companies get affected by traffic patterns especially DoorDash, Uber and even big tech companies like Google.

# Solution

By using a machine learning model to forecast traffic patterns through weather and real-time events, this would allow for better planning when traveling to locations. Next day predictions would also allow companies to optimize their employees schedules and routes based on the predicted traffic.



# Problem Description / Goal

Problem: Commuters from San Francisco waste hours on the road due to un predictable traffic patterns and relying apps like waze and google maps.

Goal:

- Build a machine learning model that can predict next-day traffic in San Francisco.
- Creating Lambda APIs for traffic patterns
- Storing model traffic patterns in a database (mostly use DynamoDB)
- Build a web application showing results of next-day traffic prediction.

# High Level Plan

Week 1-2 - Data Processing

- Analyze the dataset and its features to uncover prediction patterns

Week 3-5 - Building the model

- Build and train Gradient Boosting Model
- Train the XGBoost model on PEMS-SF dataset
- Evaluate model performance using accuracy metrics

Week 5-7 - Building website

- Using React and CSS to design a user friendly/clean website with 3 pages (About Us, Predictions, Results)

Week 8-10 - Implementing backend

- Build FastAPI server that loads the trained Gradient Boosting model
- Create REST API endpoint POST /predict to return next-day traffic
- Connect this to React frontend to allow the API to call for live predictions

# Project Evaluation Metrics

- Having accuracy as over 85% correct next-day predictions
- MAE (Mean Absolute Error) to be after about 10-15 minutes.

# Tech Stack - Desired Member Experience

- Web Development - React.js, CSS
- Backend - FastAPI, DynamoDB
- Model training - Gradient Boosting
- Data Processing - Pandas, Scikit-learn, NumPy

Will take anyone who's willing to learn and interested in this project.

# Connect

**Connect With Me!**

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